

LOGISTICS OPERATIONS ANALYSIS REPORT

2022 – 2024

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1. Executive Summary

This report presents an analysis of logistics and trucking company's operations from 2022 to 2024. The dataset contains 85,410 loads, 196,442 fuel purchases, 170,820 delivery events, and 2,920 maintenance records across 120 trucks, 150 drivers, and 200 customers.

Four areas were analyzed: customer performance, route profitability, fleet maintenance, and driver safety. The analysis reveals flat revenue of \$88 million per year, a 45% late delivery rate, rising maintenance costs from an aging fleet, and active high-risk drivers creating financial liability.

1.1. Problem Statement

The company has accumulated three years of operational data without a structured framework to evaluate performance. The key problems identified are:

- Revenue has been flat at \$88M annually for three years with no measurable growth.
- Two top revenue customers became inactive, representing over \$6M in lost annual revenue.
- Fleet maintenance costs are rising, driven by trucks manufactured as far back as 2015.
- A 45% late delivery rate points to a systemic operational failure affecting all customers.
- Active drivers with documented at-fault incident histories continue operating without intervention.

1.2. Problem Question

This analysis was conducted to answer the following business questions:

- Which customers generate the most revenue and which customer types deliver the highest value per load?
- Which routes are most profitable on a revenue-per-mile basis?
- Which trucks are consuming the most maintenance budget and generating the most downtime?
- Which drivers represent the greatest safety and financial risk?
- What are the revenue trends across the three-year period?

2. Tools and Methodology

2.1 Tools Used

PostgreSQL	Stored all 14 tables and executed analytical queries
SQL	Queried and aggregated data into 7 analytical views
Power BI	Connected to PostgreSQL views to build the interactive dashboard
pgAdmin 4	Interface for writing and running SQL queries

2.2 Process

- Loaded all 14 CSV files into PostgreSQL with proper data types and foreign key relationships.
- Assessed data quality — identified and documented null values in trips and safety tables.
- Wrote 7 SQL analytical views covering all four analysis areas.
- Connected Power BI to PostgreSQL using Import mode and built a 4-page interactive dashboard.

3. Data Overview

The dataset contains 14 tables covering the full logistics operation from January 2022 to December 2024.

Customers	200 customers — Contract, Spot, and Dedicated types
Loads	85,410 shipments with revenue and charge data
Trips	85,410 trips with driver, truck, distance, and fuel data
Drivers	150 drivers with experience and employment status
Trucks	120 trucks across Peterbilt, Freightliner, Mack, Volvo, International
Routes	58 defined routes with origin, destination, and base rates
Fuel Purchases	196,442 individual fuel transactions
Maintenance Records	2,920 events across 7 maintenance categories
Delivery Events	170,820 pickup and delivery records with on-time data
Safety Incidents	170 incidents with fault, injury, and claim data

4. Analysis and Findings

4.1 Customer Analysis

- Contract customers generate the highest average revenue per load at \$46K, compared to \$40K for Spot and \$33K for Dedicated.
- 83.77% of customers are active. XYZ Foods (\$3.09M) and Metro Group (\$3.04M) are inactive — over \$6M in lost annual revenue.
- Average detention time is 93 minutes per delivery across all customers — consistently high, indicating a company-wide scheduling problem.
- 45% of all deliveries were recorded as late, affecting the entire customer base.
- First Group generated the highest accessorial charges at \$209K.

4.2 Route Profitability

- Columbus to Los Angeles is the highest revenue-per-mile route at \$2.65, followed by Phoenix to Philadelphia at \$2.63.
- Revenue per mile across the top routes' ranges narrowly from \$2.32 to \$2.65 — limited pricing differentiation across the network.
- Average idle time is 7 hours per trip, representing driver downtime that generates no revenue.
- Average fleet fuel efficiency is 6.5 MPG, consistent across all routes.

4.3 Fleet Maintenance

- TRK00003 (2018 Peterbilt) is the most expensive truck to maintain at \$90,161 with 1,133 hours of downtime — 47 days out of service.
- The top 10 most expensive trucks are mostly 2015 models, confirming fleet age as the primary cost driver.
- Preventive maintenance (\$963K) and Repair (\$945K) are nearly equal in cost and frequency, suggesting the preventive program is not preventing breakdowns.
- Peterbilt trucks account for the highest fleet-wide downtime at 14,100 hours.

4.4 Driver Safety

- 53.85% of all incidents were at-fault — more than half caused by the company's own drivers.
- 33 incidents resulted in injuries, creating legal and insurance exposure.
- David Miller — 7 incidents, all his fault, \$118K in claims, 3 involving injuries. Still active.
- John Thomas has the highest total claims at \$132K.
- Robert Moore has \$105K in claims with zero at-fault incidents — consistently a victim of other drivers.
- Accidents (59) and Customer Complaints (51) are the most common incident types.

5. Summary of Key Findings

Analysis Area	Key Finding	Recommendation
Customer	45% late delivery rate. \$6M+ revenue lost from 2 inactive customers.	Re-engage inactive customers. Fix delivery scheduling.
Route	Revenue per mile is narrow (\$2.32–\$2.65). 7 hours idle per trip.	Review low-margin routes. Reduce idle time.
Maintenance	2015 trucks driving highest costs. Peterbilt leads downtime at 14.1K hrs.	Phase out aging fleet. Review PM schedule.
Safety	David Miller: 7 at-fault incidents, \$118K claims. Still active.	Immediate safety review. Implement driver risk scoring.
Revenue	\$88M flat for 3 years with a dip to \$87M in 2023.	Develop growth strategy. Focus on Contract customers.

6. Recommendations

6.1 Customer

- Re-engage XYZ Foods and Metro Group to recover over \$6M in lost annual revenue.
- Prioritize Contract customer acquisition — they generate \$13K more per load than Dedicated.
- Investigate the root cause of the 45% late delivery rate and implement corrective scheduling.
- Introduce detention charges for customers who consistently exceed agreed dock times.

6.2 Route

- Review routes with revenue per mile below \$2.40 for renegotiation or elimination.
- Implement appointment scheduling to reduce the 7-hour average idle time per trip.

6.3 Fleet

- Develop a phased replacement plan for 2015 and older trucks which consistently drive the highest costs.
- Conduct a repair-versus-replace analysis on TRK00003 — \$90K in costs and 47 days downtime.
- Review and increase preventive maintenance frequency for high-mileage trucks.

6.4 Safety

- Immediately review David Miller's employment status given 7 at-fault incidents and \$118K in claims.
- Implement a formal driver safety scoring system with clear intervention thresholds.
- Address Customer Complaints (51 incidents) through driver training on customer interaction and delivery procedures.
- Strengthen DOT compliance training to reduce the 27 recorded violations.

7. Conclusion

This analysis identifies clear and actionable opportunities to improve the company's financial performance and operational efficiency. The business is stable but not growing and is carrying significant hidden costs in the form of an aging fleet, poor delivery reliability, and high-risk active drivers.

The data points directly to where action is needed: reactivate lost customers, replace aging trucks, fix the delivery scheduling process, and address driver safety liabilities. The Power BI dashboard provides an ongoing tool to monitor progress across all four areas.

8. Appendix

8.1 Project Files

Database Setup	Logistics_Operations_Database_Setup.sql
Analysis Views	Logistics_Operations_Analysis_Views.sql
Dashboard	Logistics_Operations_Dashboard.pbix
Dashboard Export	Logistics_Operations_Dashboard.pdf
This Report	Logistics_Operations_Analysis_Report.docx

8.2 SQL Views

vw_customer_revenue	Customer revenue ranking with loads, revenue, and charges
vw_customer_operations	Delivery performance — detention and on-time rates per customer
vw_route_profitability	Revenue per mile, fuel efficiency, and idle time by route
vw_truck_maintenance	Individual truck maintenance costs, downtime, and repair type
vw_fleet_maintenance_by_type	Fleet-wide maintenance spending by category
vw_driver_safety	Driver incidents, fault attribution, claims, and injury data
vw_monthly_revenue_trend	Monthly revenue and load volume across 2022–2024